

How I treat plasma cell leukemia

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Plasma cell leukemia (PCL) represents an exceptionally aggressive plasma cell malignancy defined by $\geq 5\%$ circulating plasma cells in the peripheral blood of patients otherwise meeting the diagnostic criteria for multiple myeloma (MM), per the International Myeloma Working Group consensus. This ultrahigh-risk disease exhibits distinctive clinical features including frequent extramedullary involvement, severe cytopenias, hypercalcemia, renal insufficiency, and/or significantly elevated $\beta 2$ -microglobulin and lactate dehydrogenase levels. The molecular landscape includes high-risk cytogenetic abnormalities and mutations that promote accelerated proliferation, apoptotic resistance, immune evasion, and bone marrow microenvironmental independence through dysregulated adhesion molecule and chemokine receptor expression. Although autologous stem cell transplantation, proteasome inhibitors, immunomodulatory drugs, and monoclonal antibodies improved historical outcomes, the therapeutic paradigm continues to evolve. Novel therapeutic approaches including B-cell maturation antigen-directed therapies (bispecific antibodies and chimeric antigen receptor T-cell therapy), G protein-coupled receptor class C group 5 member D (GPCR5D)-targeted therapy, and B-cell lymphoma 2 (BCL-2) inhibition demonstrate promise in treating both primary and secondary PCL. Despite these advances, PCL remains inadequately studied, with treatment approaches predominantly extrapolated from MM trials in which patients with PCL have largely been excluded. This review synthesizes current evidence and presents illustrative clinical cases demonstrating practical treatment approaches, while highlighting critical knowledge gaps requiring dedicated prospective clinical trials to meaningfully improve outcomes in this challenging disease entity.

Introduction

Plasma cell leukemia (PCL) represents the most aggressive form of plasma cell dyscrasia, characterized by the presence of an elevated burden of clonal plasma cells in peripheral blood.^{1,2} PCL is classified as primary (pPCL) when presenting de novo in patients without evidence of previous multiple myeloma (MM), and secondary (sPCL) when occurring as leukemic transformation of relapsed or refractory MM. Approximately 60% to 70% of PCL cases are primary, with the remaining 30% to 40% representing sPCL.^{3,4} However, with the improvement of MM outcomes, there has been an increment in cases of sPCL.⁵

The diagnostic criteria for PCL have evolved significantly since their establishment by Kyle et al in 1974, which required both $>20\%$ circulating plasma cells and an absolute count of $>2 \times 10^9/L$ in peripheral blood.^{3,6} Retrospective studies examining the prognostic impact of various circulating plasma cell thresholds showed that patients with $\geq 5\%$ circulating plasma cells had clinical outcomes and disease characteristics remarkably similar to those meeting traditional PCL criteria.⁷ Consequently, the International Myeloma Working Group revised the diagnostic criteria in 2021, establishing that pPCL

should be defined by $\geq 5\%$ circulating plasma cells in the peripheral blood of patients otherwise meeting the diagnostic criteria for MM.⁸

Circulating tumor cells (CTCs) levels carry adverse prognostic significance on a continuum, with relevant thresholds as low as $\geq 2\%$ CTCs exhibit distinct molecular features vs intramedullary plasma cells, including loss of adhesion molecules enabling bone marrow (BM) independence and altered chemokine receptor expression promoting peripheral circulation and extramedullary tropism.⁹

The prognosis of PCL remains poor despite therapeutic advances, with historical median overall survival (OS) of only 7 months with conventional chemotherapy.¹⁰ Recent multicenter data demonstrates a median OS of 36.6 months for pPCL and only 3.2 months for sPCL, with significantly improved outcomes for patients who received autologous stem cell transplantation (ASCT) in both subtypes (49.5 months vs 12.5 months for pPCL and 16 months vs 3.4 months for sPCL).¹¹ This review examines the current understanding of PCL biology, clinical presentation, and therapeutic strategies, with emphasis on emerging treatment paradigms and the critical need for dedicated clinical trials in this ultrahigh-risk population.

CASE 1: pPCL IN A YOUNG PATIENT WHO INITIALLY OPTED AGAINST CONVENTIONAL CHEMOTHERAPY

A 50-year-old man presented with fatigue and back pain. Laboratory tests showed the following values: hemoglobin (Hgb), 9.2 g/dL; creatinine (Cr), 1.4 mg/dL; immunoglobulin G κ M-protein, 4.5 g/dL; and β 2-microglobulin, 5.2 mg/L. BM biopsy showed 70% atypical plasma cells. Peripheral blood showed a circulating plasma cell count of $4.5 \times 10^9/L$, accounting for 22% of the differential white cell count. Fluorescence in situ hybridization (FISH) reported del(17p), dup1q, and t(4;14). Diffusion-weighted magnetic resonance imaging showed multiple lytic lesions. The patient was diagnosed with pPCL.

The patient received Dara-KRd (daratumumab and hyaluronidase-fihj [Darzalex Faspro], carfilzomib [Kymprolis], lenalidomide, and dexamethasone) for 8 cycles achieving very good partial response. Despite high-risk features and recommendation to proceed with ASCT, he declined up-front transplantation but underwent stem cell harvest and cryopreservation. He continued Dara-KRd for 6 additional cycles and then transitioned to KRd maintenance. After 14 months, he developed relapsed disease. VPd (bortezomib, pomalidomide, and dexamethasone) salvage showed no response. Given the refractory disease, he proceeded to high-dose melphalan (MEL200) ASCT.

On day +60 post-ASCT, he developed progression and new extramedullary disease (EMD). He was evaluated for chimeric antigen receptor (CAR) T-cell therapy and underwent leukapheresis for ciltacabtagene autoleucel (cilta-cel) manufacturing. He received 1 cycle of VD-PACE (bortezomib, dexamethasone, cisplatin, doxorubicin, cyclophosphamide, etoposide) as bridging chemotherapy during the 6-week manufacturing period. He then received cilta-cel infusion and developed grade 1 cytokine release syndrome managed with tocilizumab without immune effector cell-associated neurotoxicity syndrome.

Response was rapid with undetectable M-protein at week 4 and stringent complete response in a month. The BM at month 3 showed measurable residual disease (MRD)–negative status by next-generation flow cytometry, with a sensitivity of 10^{-5} . He remained off therapy in sustained stringent complete response with MRD-negative status at 9 months after infusion, requiring monthly IV immunoglobulin for hypogammaglobulinemia.

This case illustrates several key points in pPCL management. Although deferring transplant is generally not recommended given the significant survival benefit of early ASCT in pPCL, patient preferences may not always align with medical recommendations. Early stem cell collection preserved transplant options despite initial deferral. The rapid, deep MRD-negative response to cilta-cel after early post-ASCT relapse with high-risk cytogenetics demonstrates the unique efficacy of B-cell maturation antigen (BCMA)–directed CAR T-cell therapy even in aggressive pPCL with prior treatment resistance. Ongoing surveillance is critical given the high relapse risk in PCL, despite initial CAR T-cell response. Although maintenance therapy with BCMA bispecific antibody (BsAb) was discussed given this high-risk setting, the patient opted against this approach.

Pathobiology of PCL

The pathobiology of PCL involves a complex interplay of genetic aberrations, epigenetic modifications, altered adhesion molecule expression, immune evasion mechanisms, and independence from the BM microenvironment (Figure 1). We suggest the following testing be conducted in all patients with pPCL (Table 1).

Clinical presentation

Patients with pPCL typically present at a younger median age (60–64 years) than patients with MM (66–70 years), although they have a significantly worse performance status at diagnosis.^{12,13} Clinical manifestations reflect higher tumor burden and more aggressive disease biology, with patients with pPCL more frequently presenting with severe anemia, thrombocytopenia, hypercalcemia, and renal impairment.^{5,12–14} Compared with patients with newly diagnosed MM, patients with pPCL showed greater disease burden, with higher frequencies of hypercalcemia (24% vs 6%) and renal failure (25% vs 3%).¹⁴ Laboratory parameters reflect aggressive disease with significantly elevated β 2-microglobulin and markedly increased lactate dehydrogenase levels. EMD represents a particularly important distinguishing feature, occurring in 20% to 34% of patients with pPCL compared with only 5% to 10% of patients with newly diagnosed MM.^{11,14–16} The incidence of EMD is even higher in sPCL than in pPCL.¹¹

Peripheral blood examination typically reveals a leukoerythroblastic picture in up to 67% of cases, with circulating plasma cells displaying variable morphology ranging from mature-appearing cells to immature plasmablasts, often with binucleated forms, Russell bodies (cytoplasmic immunoglobulin inclusions), and Dutcher bodies (intranuclear inclusions).^{5,17,18} Figure 2 shows peripheral blood morphology in PCL.

Skeletal involvement in PCL presents differently from MM, with a lower incidence of extensive lytic bone lesions (59% vs 81%) likely reflecting the reduced interaction between leukemic plasma cells and the BM microenvironment.^{8,12,14} Central nervous system involvement, although rare, occurs more frequently in PCL than in MM, representing an association with extremely poor prognosis and requiring consideration of prophylactic intrathecal therapy in high-risk patients.^{19,20}

Biology

Unlike MM, in which hyperdiploidy is present in ~50% of cases and generally associated with favorable prognosis, hyperdiploidy is present in only 0% to 8.8% of pPCL cases, with hypodiploidy and nonhyperdiploid karyotypes predominating.^{5,8,14} pPCL showed significantly higher frequencies of adverse cytogenetic features compared with newly diagnosed MM, including high-risk FISH abnormalities (47% vs 18%), immunoglobulin heavy chain translocations (49% vs 26%), del1p32 (35% vs 10%), del17p13 (39% vs 10%), and t(11;14) (50% vs 19%). Multiple molecular abnormalities frequently co-occurred. Importantly, despite sharing del(17p), pPCL and newly diagnosed MM exhibit distinct transcriptomic profiles (with over 3500 differentially expressed genes and significant divergence in RNA splicing machinery), underscoring that

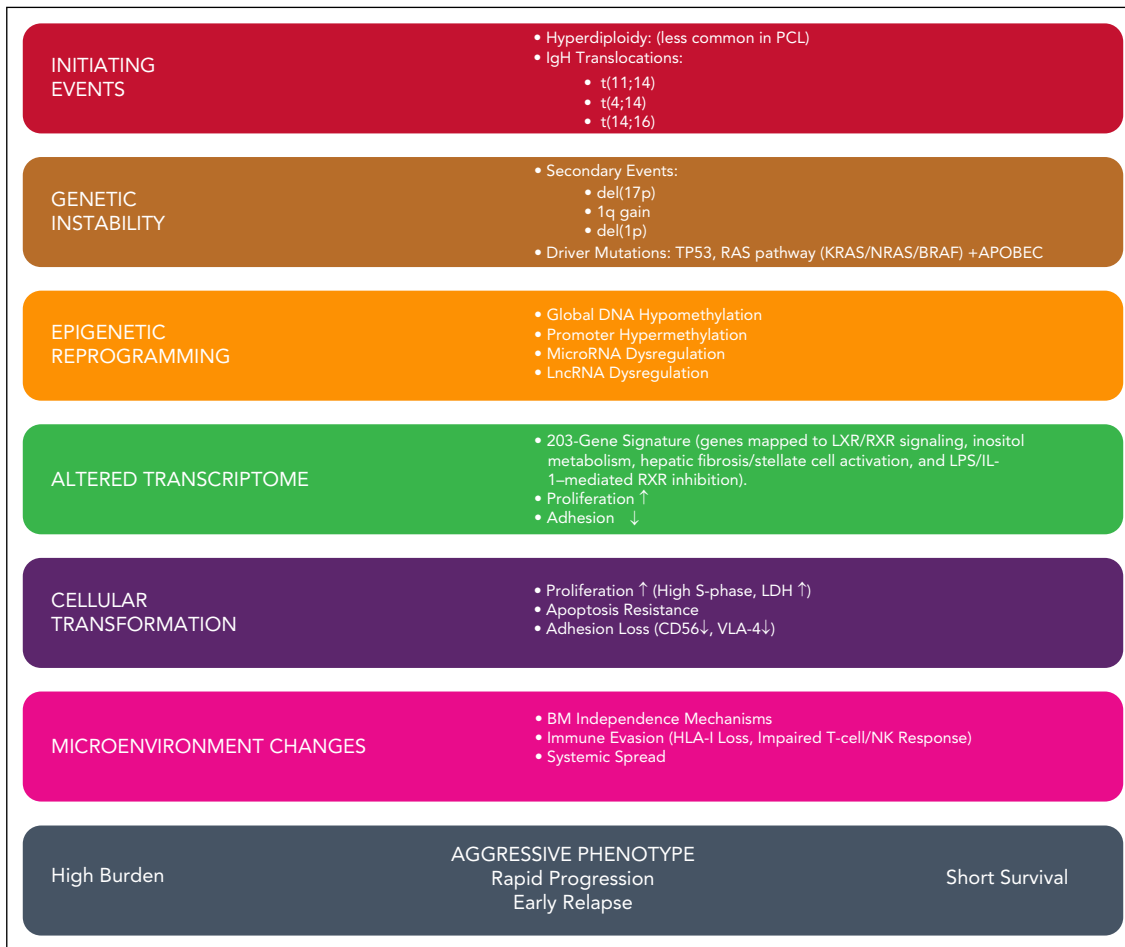


Figure 1. Pathobiology of PCL. APOBEC, apolipoprotein B mRNA editing enzyme catalytic polypeptide-like; IgH, immunoglobulin heavy chain; IL-1, interleukin 1; LDH, lactate dehydrogenase; LncRNA, long noncoding RNA; LPS, lipopolysaccharide; NK, natural killer cell; VLA, very late antigen.

cytogenetic concordance does not imply biological equivalence between these entities.²¹

The t(11;14) translocation, resulting in *CCND1* dysregulation, is notably prevalent in pPCL, occurring in 39% of patients, thereby making it the most common immunoglobulin heavy chain translocation observed.²² The t(11;14) subgroup may represent a distinct biological subset within pPCL, with favorable prognostic impact, although this is substantially abrogated when co-occurring with high-risk abnormalities. In a retrospective study comparing pPCL and sPCL, the gain/amplification of 1q21 (1q21+) was identified as the most common cytogenetic finding in sPCL, present in 75% of cases.²³

Proteomic profiling has identified upregulated CRIP1 and CRIP2 in pPCL,²⁴ whereas a “PCL-like” transcriptional signature identified within MM cohorts reinforces that PCL biology exists along a continuum rather than a binary entity.¹⁴

In a comprehensive analysis of 1474 patients enrolled in total therapy protocols, cytogenetic abnormalities were significantly overrepresented in patients with pPCL (81% vs 32% in patients without pPCL; $P < .001$). Hypodiploidy was observed in 48% of patients with pPCL vs 12% of patients without pPCL ($P < .001$).

Gene expression profiling showed that pPCL samples formed a tight, uniform cluster distinct from non-pPCL samples, with overrepresentation in the CD-1 and MF molecular subgroups (both of which are associated with high-risk disease). Importantly, 203 differentially expressed gene probes distinguished pPCL from non-pPCL, with pathway analysis identifying enrichment in genes involved in lipid metabolism pathways, including liver X receptor/retinoid X receptor (LXR/RXR) activation and hepatic stellate-cell activation.²⁵

Beyond cytogenetic abnormalities, pPCL is characterized by a distinct mutational landscape that drives genomic instability and aggressive behavior. *TP53* mutations occur in ~20% to 30% of pPCL cases, significantly higher than those in newly diagnosed MM, and are associated with particularly poor outcomes.^{26,27} RAS pathway mutations (*KRAS*, *NRAS*, and *BRAF*) are frequently observed, with *KRAS* mutations detected in up to 40% of cases, contributing to enhanced proliferation and survival signaling.^{27,28} Notably, apolipoprotein B mRNA editing enzyme catalytic polypeptide-like mutational activity has been observed in a subset of pPCL cases, potentially enriched in certain translocation contexts (eg, *MAF*-driven disease), contributing to genomic instability and clonal evolution in affected cases; however, this signature is not a universal feature of pPCL and should not be overgeneralized.²⁹ This

Table 1. Initial diagnostic workup in pPCL

Category	Required tests and evaluations
Hematology and biochemistry	Complete blood count with differential Comprehensive metabolic panel (renal function, hepatic function, and calcium) Lactate dehydrogenase level β2-Microglobulin Serum albumin Coagulation tests (PT/INR and aPTT)
Serum and urine protein studies	Serum protein electrophoresis with quantification of M-protein Serum immunofixation electrophoresis Serum free light chain assay with κ/λ ratio Quantitative immunoglobulins (IgG, IgA, and IgM) 24-h urine protein electrophoresis with quantification Urine immunofixation electrophoresis
BM evaluation	BM aspiration with morphologic assessment BM trephine biopsy (particularly if dry tap) Quantification of clonal plasma cells in the BM
Flow cytometry (immunophenotyping)	Multiparameter flow cytometry on the BM and peripheral blood plasma cells Recommended immunophenotypic panel: CD38 and CD138 (plasma cell markers) CD19, CD20, and CD27 (B-cell markers) CD45 and CD56 (lineage and adhesion markers) CD81 and CD117 (differentiation markers) Cytoplasmic κ and λ light chains (clonality assessment)
Cytogenetics and molecular testing	FISH on purified BM plasma cells (and peripheral blood if necessary): Ploidy status (hyperdiploidy vs hypodiploidy) t(11;14)(q13;q32): <i>CCND1/IGH</i> t(4;14)(p16;q32): <i>FGFR3/MMSET/IGH</i> t(14;16)(q32;q23): <i>MAF/IGH</i> del(17p): <i>TP53</i> deletion Gain/amplification of 1q21 del(1p): 1p21 deletion del(13q): monosomy 13 or 13q deletion Next-generation sequencing: emerging preferred standard if available A well-designed panel enables rapid, single-assay detection of all clinically relevant prognostic markers, including actionable alterations such as <i>BRAF</i> , as well as assessment for <i>TP53</i> mutations.
Imaging studies	Whole-body low-dose CT combined with PET (PET/CT): standard approach for detection of osteolytic lesions and EMD Whole-body MRI, particularly diffusion-weighted MRI: emerging preferred standard if available
CNS evaluation	Diagnostic lumbar puncture if neurological symptoms are present Consider prophylactic lumbar puncture in selected high-risk patients (eg, very high circulating plasma cell count), preferably after initial treatment reduces circulating plasma cells to prevent CNS seeding

aPTT, activated partial thromboplastin time; CNS, central nervous system; CT, computed tomography; IgG/A/M, immunoglobulin G/A/M; INR, international normalized ratio; MRI, magnetic resonance imaging; PET, positron emission tomography; PT, prothrombin time.

mutational burden is accompanied by profound epigenetic reprogramming, including global DNA hypomethylation coupled with promoter-specific hypermethylation affecting tumor suppressor genes.³⁰ MicroRNA (miR) profiling has revealed distinct dysregulation patterns in pPCL compared with MM, with the downregulation of the *miR-15a/16-1* cluster and overexpression of oncogenic miRs contributing to the aggressive phenotype.^{31,32}

The cellular phenotype of pPCL reflects its aggressive biological behavior, characterized by markedly elevated proliferative activity with high S-phase fractions (often >3%) and elevated lactate dehydrogenase levels, contrasting sharply with the typically quiescent nature of BM-resident myeloma cells.^{12,33} A defining feature is the loss of adhesion molecule expression, particularly CD56 (neural cell adhesion molecule [NCAM])

downregulation in 70% to 80% of cases and reduced very late antigen (VLA)-4 (CD49d) expression, which facilitates detachment from the BM microenvironment and enables autonomous growth independent of stromal support.^{34,35} This loss of BM dependence, coupled with resistance to apoptosis through the dysregulation of B-cell lymphoma 2 (BCL-2) family proteins and activation of survival pathways, allows malignant plasma cells to survive and proliferate in peripheral circulation without the protective BM niche. The autonomous growth capacity and altered cytokine dependency patterns distinguish pPCL from MM, in which malignant cells remain largely dependent on BM stromal cell interactions.

Immune evasion mechanisms play a critical role in pPCL pathogenesis and contribute to its aggressive clinical course.³⁶ These immune defects, combined with the complex structural

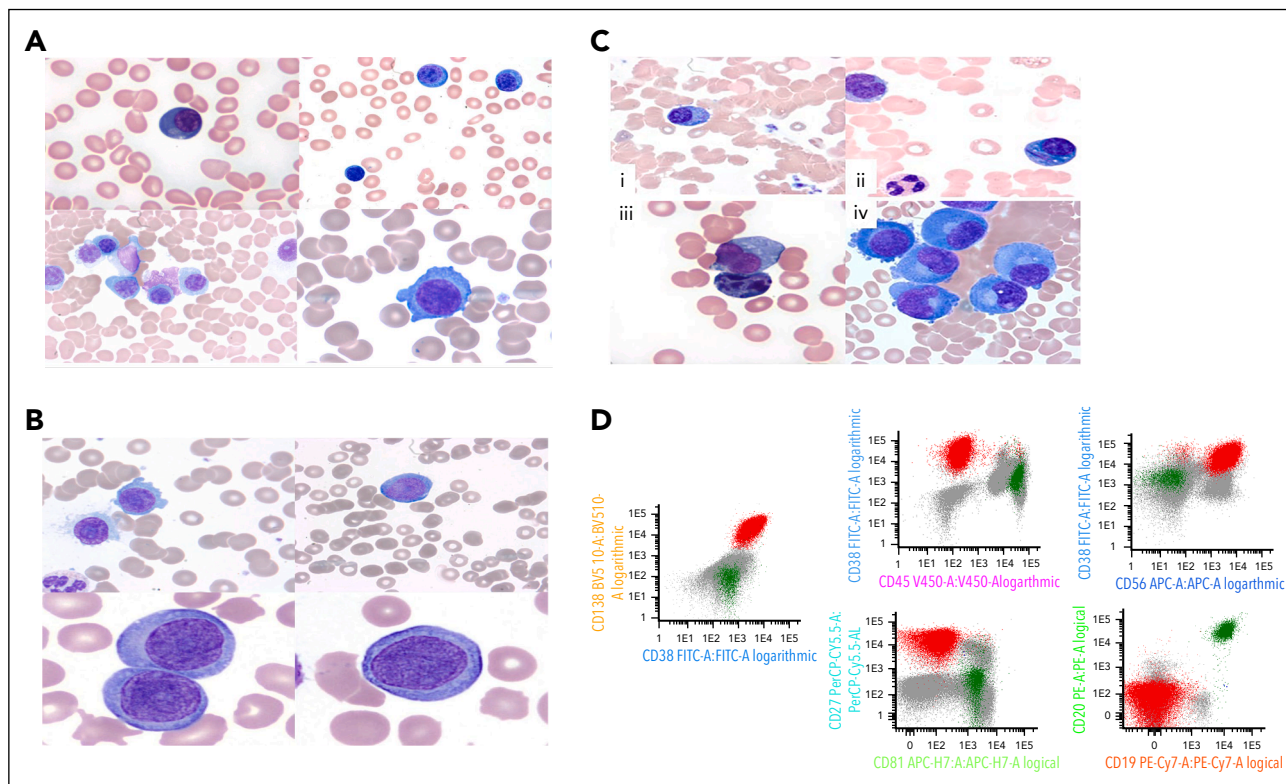


Figure 2. Peripheral blood morphology in PCL. (A) Composite panel showing circulating plasma cells demonstrating morphologic heterogeneity. Cells exhibit typical plasma cell features including eccentric nuclei with clumped chromatin, perinuclear hof, and basophilic cytoplasm, with variation in nuclear-to-cytoplasmic ratio and degree of chromatin condensation among individual cells. (Images at ×100, ×50, ×50 and ×100 magnification, respectively; Wright-Giemsa stain.) (B) Composite panel of circulating plasmablasts. These cells display immature features with less condensed chromatin, more prominent nucleoli, and higher nuclear-to-cytoplasmic ratios compared with mature plasma cells. The cells demonstrate the plasmablastic morphology characteristic of aggressive plasma cell neoplasms. (Images ×100 magnification; Wright-Giemsa stain.) (C) Representative morphologic variants in PCL. (i-ii) Circulating IgA-secreting plasma cells ("flame cells") showing characteristic bright eosinophilic or flame-colored cytoplasm due to IgA accumulation. (iii) Binucleated plasma cell demonstrating multinucleation, a common feature in plasma cell dyscrasias. (iv) Circulating plasmablasts with cytoplasmic blebs, indicating active membrane remodeling and potential markers of cellular stress or migration. (First image, ×50 magnification; remaining images, ×100 magnification; Wright-Giemsa stain). (D) Immunophenotypic analysis of leukemic plasma cells (highlighted in red), compared to background B-cells (green). The leukemic plasma cells are positive for CD138, CD38, CD56, and CD27, whereas negative for CD45, CD19, and CD81. Cells were also negative for CD20 (not shown). APC, allophycocyanin; FITC, fluorescein isothiocyanate; hof, description of the area of pallor by the nucleus; IgA, immunoglobulin A; PE, phycoerythrin.

alterations, high mutational burden (including high rates of biallelic *TP53* inactivation, particularly in t(11;14) patients), and epigenetic dysregulation characteristic of pPCL,³⁷ create a permissive environment for rapid disease progression and early treatment resistance, ultimately contributing to the dismal prognosis associated with this aggressive disease.

Prognostic models

The prognosis of PCL remains poor. A multicenter study developed a risk stratification model identifying age ≥60 years, platelet count of ≤100 × 10⁹/L, and peripheral plasma cell count of ≥20 × 10⁹/L as independent adverse predictors, with median survival ranging from 46 months in patients with no risk factors to just 12 months in those with 2 or 3 factors.³⁸ However, emerging data suggest molecular heterogeneity may influence outcomes (pPCL with t(11;14) appears to confer better prognosis [median OS, 39 vs 18 months] when occurring in isolation without concurrent high-risk features,^{22,39} whereas non-t(11;14) pPCL is characterized by biallelic *TP53* alterations, distinct *IRF4* mutations, and a divergent transcriptomic profile),³⁹ highlighting the need for refined prognostic models that account for specific genetic profiles (Table 2). sPCL carries a worse prognosis than pPCL, with historical median survival of

only 2 to 7 months, reflecting both aggressive leukemic transformation and treatment resistance from prior multiagent exposure.¹¹

CASE 2: pPCL IN AN OLDER PATIENT FIT FOR INTENSIVE CHEMOTHERAPY

A 72-year-old woman presented with progressive fatigue and worsening back pain. Laboratory tests showed the following values: Hgb, 9.2 g/dL; white blood cell count, 3.4 × 10⁹/L; platelet count, 53 × 10⁹/L; Cr, 1.0 mg/dL; albumin, 3.5 g/dL; and calcium, 8.4 mg/dL. Serum protein electrophoresis showed no monoclonal protein with normal free light chain levels in the serum. BM biopsy revealed 69% atypical plasma cells. Peripheral blood showed 12% circulating plasma cells. FISH showed del(13q) and t(11;14). The patient was diagnosed with pPCL. She initially received 1 cycle of Dara-Rd locally with no response and was transferred to our institution. She then received 2 cycles of VD-PACE achieving PR. Following response, she underwent tandem ASCT with MEL200 conditioning for both transplants. She remained in remission for over 1 year after transplant on RVD (lenalidomide, bortezomib, and dexamethasone) maintenance therapy. At first relapse,

Table 2. Summary of prognostic factors in PCL

Risk category	Prognostic factors
Adverse prognostic factors	Advanced age (>60-65 y) High absolute number of circulating plasma cells Thrombocytopenia Presence of extramedullary plasmacytomas Elevated β 2-microglobulin level Elevated lactate dehydrogenase level High-risk cytogenetics: del(17p)/TP53 mutations t(4;14) t(14;16) Gain/amplification 1q21 del(1p) Hypodiploidy Complex karyotype Multiple concurrent high-risk abnormalities (≥ 2)
Favorable prognostic factors	t(11;14)(q13;q32): particularly when present in isolation without other high-risk features Note: favorable prognosis is significantly reduced if t(11;14) occurs with concurrent high-risk cytogenetic abnormalities

she received KPd (carfilzomib, pomalidomide, and dexamethasone) achieving PR. At second relapse, she received venetoclax-based therapy with initial response followed by progression. She subsequently developed sPCL and progressed through teclistamab and then talquetamab, both of which provided brief disease control of <3 months.

The patient was evaluated for CAR T-cell therapy. She underwent leukapheresis for cilta-cel manufacturing and received bridging chemotherapy during the manufacturing period. Following cilta-cel infusion, she initially showed disease response. However, at day +40 postinfusion, with the recurrence of leukemic phase disease, presenting as sPCL after having achieved intramedullary disease control for over 3 years since initial pPCL diagnosis. Despite supportive care, she died from progressive disease at day +40 after cilta-cel. OS from initial pPCL diagnosis was over 3 years.

This case illustrates how initial disease biology can govern long-term outcomes despite achieving deep responses with multiple therapies. Although VD-PACE followed by tandem ASCT provided over 1 year of remission, the pattern of frequent relapses with progressively shorter durations of response to subsequent lines, including novel agents, BsAbs, and ultimately CAR T-cell therapy, reflected the inherently aggressive nature of her disease. This case underscores that despite the availability of multiple effective treatment modalities, initial disease characteristics, can predict ultimate outcomes and emphasize the need for novel approaches in this high-risk population.

Management of pPCL

Pretreatment assessment and supportive care

Comprehensive pretreatment evaluation and proactive supportive care are essential for successful PCL management (Figure 3). After the completion of diagnostic testing (Table 1), patient fitness should be assessed using the International Myeloma Working Group frailty score and performance status criteria to determine eligibility for intensive therapy including

ASCT.⁴⁰ Treatment should begin promptly after diagnosis confirmation. Supportive care interventions must be instituted immediately, with particular emphasis on tumor lysis syndrome prevention through aggressive IV hydration and uric acid reduction.⁴¹ Most patients require close monitoring in an inpatient setting during initial treatment to manage acute complications including hypercalcemia, infections, and cytopenias. Additional supportive measures include risk-stratified thrombosis prophylaxis, antimicrobial prophylaxis, and skeletal protection with monthly bone-modifying agents.⁴²⁻⁴⁴

Fit patients For fit patients with pPCL, an intensive treatment approach incorporating novel agents, ASCT, and prolonged maintenance therapy has emerged as the standard of care, with accumulating evidence demonstrating meaningful improvements in survival outcomes compared with historical series^{25,45-57} (Figure 4; Table 3).

We recommend initiating therapy with quadruplet regimens for 4 to 6 cycles as induction in patients not requiring immediate hospitalization. The use of cytoreduction with VD-PACE chemotherapy for 1 or 2 cycles is recommended for patients who may require immediate treatment while hospitalized, given that most patients with pPCL often present with cytopenias that preclude standard outpatient therapy for the first cycle. Based on recent data showing superior outcomes with anti-CD38 antibody-containing regimens, daratumumab-based combinations such as D-VRd (daratumumab, bortezomib, lenalidomide, and dexamethasone) or isatuximab-based quadruplet regimens are preferred when available.⁵⁸⁻⁶³ Using carfilzomib instead of bortezomib is preferred when accessible, given its superior efficacy profile in high-risk disease.^{62,64,65} The OPTIMUM trial specifically evaluated an intensive D-CVRd (daratumumab, cyclophosphamide, bortezomib, lenalidomide, and dexamethasone) quintuplet regimen in 107 patients with ultrahigh-risk myeloma and pPCL (n = 10), showing a 30-month progression-free survival (PFS) of 77% and median PFS not yet reached after more than 3 years of follow-up.⁴⁷

Following induction, fit patients should proceed to tandem ASCT using high-dose melphalan (200 mg/m²) conditioning

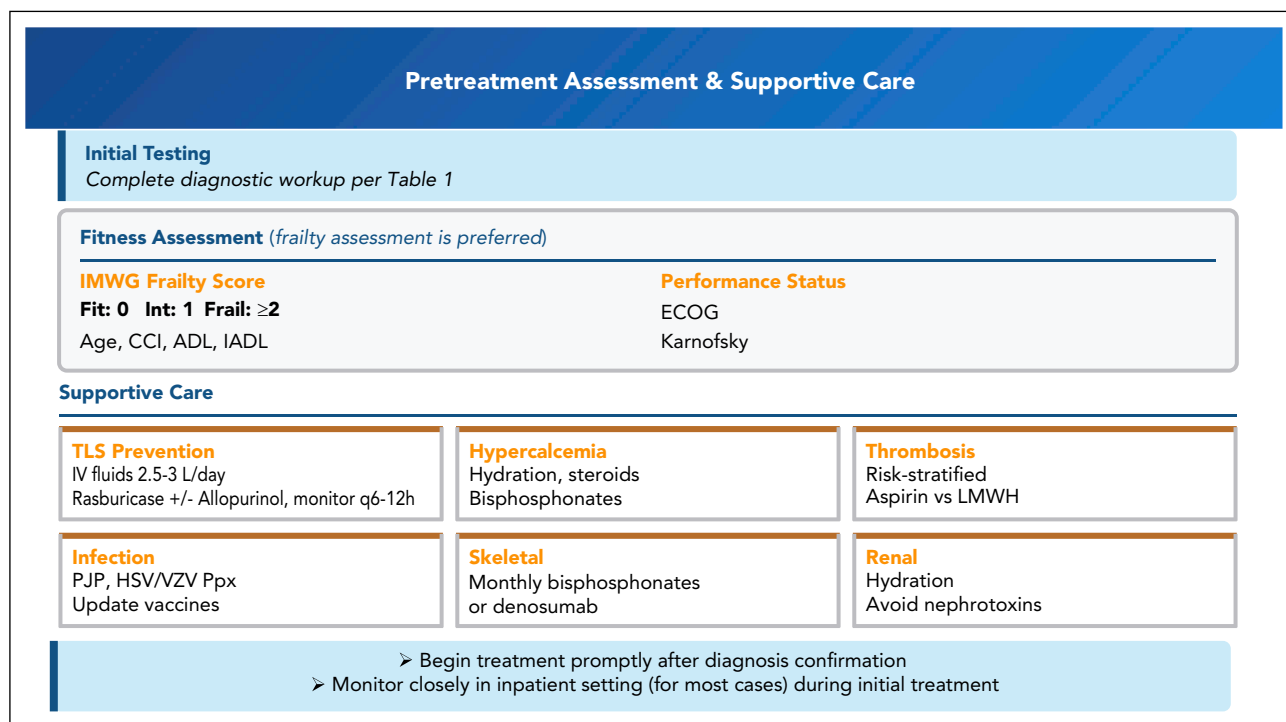


Figure 3. Pretreatment assessment and supportive care. ADL, activities of daily living; CCI, Charlson Comorbidity Index; ECOG, Eastern Cooperative Oncology Group; HCT, hematopoietic cell transplantation; HSV, herpes simplex virus; IADL, instrumental activities of daily living; IMWG, International Myeloma Working Group; Int, Intermediate; LMWH, low-molecular-weight heparin; PJP, *Pneumocystis jirovecii* pneumonia; Ppx, prophylaxis; q6-12h, every 6 to 12 hours; TLS, tumor lysis syndrome; VZV, varicella zoster virus.

when feasible; otherwise, patients should proceed with at least single ASCT following induction treatment.^{62,64,66,67} We do not recommend delaying ASCT based on improved outcomes, with the use of ASCT in pPCL demonstrated in multiple older retrospective studies.¹¹ The European Myeloma Network

(EMN) EMN12/HOVON-129 trial showed that this approach in younger patients (age of 18-65 years) achieved an overall response rate of 86%, with 50% achieving complete remission or better. Among the 20 patients evaluated for MRD, 80% achieved MRD negativity. The median PFS was 15.5 months

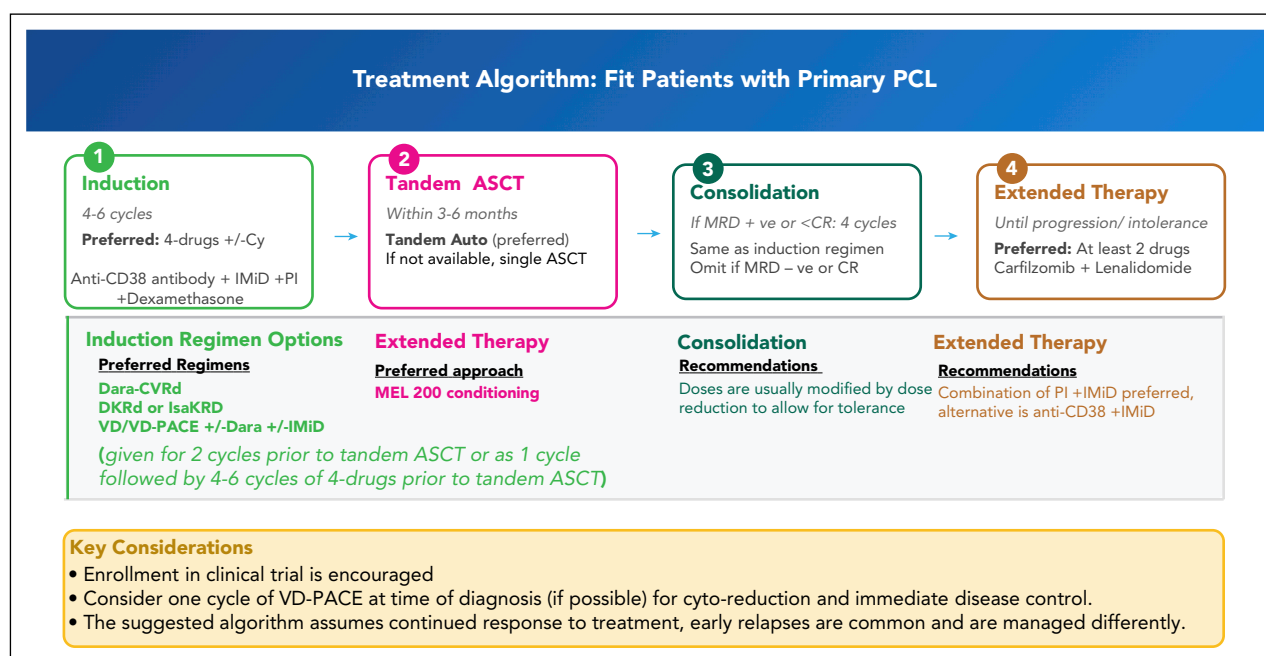


Figure 4. Treatment approach for pPCL. CR, complete response; Cy, cyclophosphamide; IMiD, immunomodulatory drug; PI, proteasome inhibitor; IsaKRd, isatuximab, carfilzomib, lenalidomide, and dexamethasone; VD/VD-PACE, bortezomib, dexamethasone with/without cisplatin, doxorubicin, cyclophosphamide, and etoposide; +ve, positive; -ve, negative.

Table 3. Prospective studies in pPCL

Study	Regimen	Definition of PCL	No. of patients	Median follow-up, mo	ORR	Median PFS, mo	Median OS, mo
Musto et al ⁴⁸	Rd	≥20% CPCs	23 (15 TE and 8 TIE)	34	74%	14	28
Royer et al ⁵⁰	PACE-D + ASCT	>2 × 10 ⁹ /L CPCs or >20%	40	28.7	69%	15.1	36.3
Van de Donk et al ⁵¹ (EMN12/HOVON-129)	KRd ± ASCT	>2 × 10 ⁹ /L CPCs or >20%	36 (age 18-65 y)	43.5	83% (end of induction)	15.5	28.4
			25 (age ≥66 y)	32	80% (end of induction)	13.8	24.8
Yan et al ⁹⁵ (CAREMM-002)	BCMA CAR T-cells	≥5% CPCs	8 (evaluable)	15.4	100% (87.5% sCR)	NR (longest documented duration of remission is 26.7 mo)	NR
Xu et al ⁹³ (CART sandwich)	CAR T-cell-ASCT-CAR T-cell	≥5% CPCs	20	8.0	95% MRD-neg	NR	NR*
Touzeau et al ⁶² (IFM 2018-04)	Dara-KRd + tandem ASCT	High-risk MM + PCL (≥5% CPCs)	50 (4 PCL)	33	95% (end of induction)	30 mo: 80%	30 mo: 91%
Kaiser et al ⁴⁷ (UK OPTIMUM/MUKnine)	Dara-CVRd + ASCT	PCL (>20% CPCs) or UHR MM	107 (10 PCL)	41.2	94%†	30 mo: 77%	30 mo: 83.5%
Usmani et al (SWOG-1211) ⁵⁷	RVd ± Elo	≥2000 CPCs per mL or >20% on differential	100 (7 PCL)	53	85% (RVd)/83% (RVd-Elo)†	33.6 (RVd)/31.5 (RVd-Elo)	NR (RVd)/68 (RVd-Elo)

CPCs, circulating plasma cells; Dara, daratumumab; Elo, elotuzumab; MRD-neg, MRD-negative; NR, not reached/not reported; ORR, overall response rate; Rd, lenalidomide and dexamethasone; sCR, stringent complete response; TE, transplant-eligible; TIE, transplant-ineligible; UHR MM, ultrahigh-risk MM.

*Deaths in the CART sandwich trial were due to adenovirus infection after first CAR T-cell infusion.

†IFM 2018-04, OPTIMUM, and SWOG 1211 trials included patients with PCL as part of the high-risk myeloma cohorts; exact PCL-specific outcomes are not separately reported in detail.

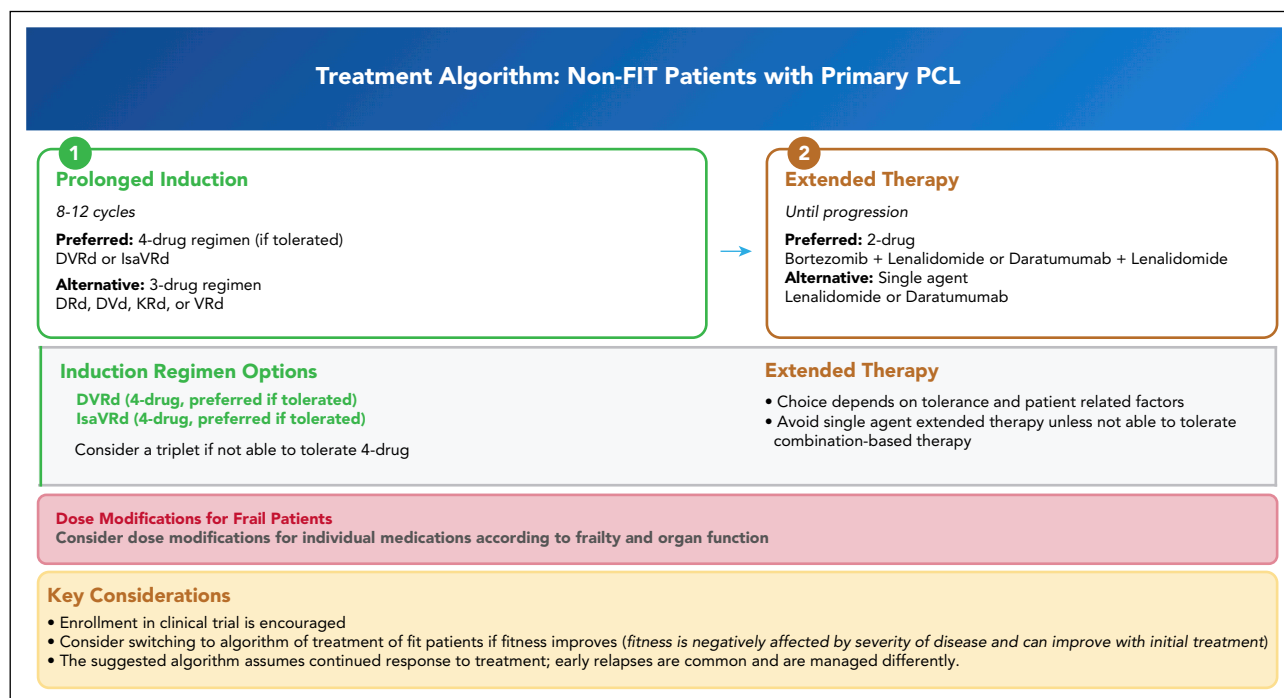


Figure 5. Treatment approach for sPCL. CR, complete response; IMiD, immunomodulatory drug; PI, proteasome inhibitor.

and OS was 28.4 months.⁵¹ Another prospective study evaluating tandem ASCT with Dara-KRd induction and consolidation in high-risk newly diagnosed MM enrolled 50 patients. Among 3 patients with pPCL, 1 had early disease progression during induction, whereas the other 2 patients remained progression free in the maintenance phase. The 30-month PFS and OS for all patients were 80% and 91%, respectively.⁶² Large retrospective analyses have also confirmed the benefit of ASCT in pPCL. The Center for International Blood and Marrow Transplant Research study of 97 patients showed 3-year PFS and OS of 34% and 64%, respectively, with acceptable treatment-related mortality of 5%.⁵⁵

Extended consolidation and maintenance therapy are critical components for maximizing the depth of response and prolonging remission. We recommend 4 cycles of quadruplet-based consolidation therapy following ASCT when possible, followed by continuous maintenance with lenalidomide combined with either a proteasome inhibitor (carfilzomib preferred) or an anti-CD38 antibody until disease progression or intolerable toxicity. This strategy utilizes the proven long-term efficacy of proteasome inhibitor–lenalidomide or anti-CD38 antibody–lenalidomide combinations for sustained disease control. Importantly, this approach preserves therapeutic options at relapse: patients who discontinue anti-CD38 antibody maintenance without developing refractory disease can be re-treated with alternative anti-CD38 antibody-based combinations.⁶⁸

The rationale for extended maintenance in pPCL is intriguing. The EMN12/HOVON-129 trial used KR (carfilzomib–lenalidomide) maintenance following consolidation, with patients receiving maintenance for a median of 26.6 months in the younger cohort.⁵¹ This approach addresses the fundamental biology of pPCL, in which MRD persistence leads to early

relapse.⁴⁷ The achievement of MRD-negativity, particularly in high-risk cytogenetics, is strongly associated with improved outcomes, and maintenance therapy provides the best opportunity to achieve and sustain this status.⁶⁹

MRD assessment in PCL should combine BM evaluation via a validated high-sensitivity technique (next-generation flow cytometry or next-generation sequencing, sensitivity $\geq 10^{-5}$) with functional imaging (PET-CT and diffusion-weighted MRI) to detect EMD not captured by marrow sampling alone. Modality selection should be guided by local laboratory availability and expertise. We recommend initiating consolidation/maintenance therapy early after ASCT ideally by day +60 if adequate count recovery has been achieved given the high risk of rapid early progression in pPCL.

For highly selected young patients (age <65 years) with an available matched donor who did not achieve complete response after autologous ASCT and with limited access to therapeutic options, reduced-intensity conditioning allogeneic SCT may be considered.^{55,70,71} However, we do not generally recommend allogeneic SCT given the high treatment-related mortality and limited evidence of graft-vs-myeloma benefit in PCL, particularly when other therapies are available.^{72,73}

Nonfit patients For patients with pPCL who are not fit for intensive therapy and/or ASCT due to advanced age, comorbidities, or frailty, treatment outcomes remain challenging, although the introduction of novel agents has improved survival compared with historical series with conventional chemotherapy.⁷⁴⁻⁷⁶

We recommend prolonged induction therapy for 8 to 12 cycles with quadruplet regimens when tolerated. (Figure 5) Preferred regimens include daratumumab-based quadruplets such as

D-VRd isatuximab-based combinations (IsaVRd). For patients unable to tolerate 4-drug regimens, alternative triplet options include DRd (daratumumab, lenalidomide, dexamethasone), DVd (daratumumab, bortezomib, dexamethasone), KRd (carfilzomib, lenalidomide, dexamethasone), or VRd. The EMN12/HOVON-129 trial demonstrated that older patients (age ≥ 66 years) treated with KRd induction followed by KR maintenance achieved a median PFS of 13.8 months and OS of 24.8 months, representing a doubling of survival compared with previous retrospective series.⁵¹

Following induction, we recommend extended therapy until disease progression with 2-drug maintenance combining bortezomib or carfilzomib with lenalidomide or daratumumab with lenalidomide. The choice depends on tolerability and patient-related factors, with the goal of maintaining disease control while preserving quality of life. Single-agent maintenance with lenalidomide or daratumumab may be considered as an alternative for frail patients who cannot tolerate combination therapy. Importantly, we recommend avoiding single-agent extended therapy unless patients are unable to tolerate combination-based approaches, as dual-agent maintenance provides superior disease control in high-risk patients.⁷⁴⁻⁷⁶

Dose modifications should be implemented according to frailty and organ function, managing cytopenias, neuropathy, and infections. Frail patients may require further reductions and schedule modifications to maintain tolerability while optimizing control.^{40,77}

CASE 3: pPCL REFRACTORY TO UP-FRONT CONVENTIONAL CHEMOTHERAPY

A 58-year-old man presented with primary refractory disease. He developed back pain and was found to have a T12 fracture. Laboratory tests showed the following values: Hgb, 5.9 g/dL; white blood cell count, $7.78 \times 10^9/L$; platelet count, $13 \times 10^9/L$; Cr, 3.50 mg/dL; calcium, 10.8 mg/dL; and albumin, 3.6 g/dL. Serum protein electrophoresis showed κ light chain monoclonal protein. Serum free light chains showed κ of 67.5 mg/dL and λ of 0.8 mg/dL, with a κ/λ ratio of 81.5. BM biopsy showed 95% plasma cells and more than 22% circulating plasma cells. The patient was diagnosed with pPCL. He initially received 2 cycles of VD-PACE at a different institution with progression of his disease followed by 1 cycle of D-VRd with no response and was referred to our institution.

He was admitted with repeat BM showing 100% plasma cells, and FISH showed t(11;14). He was started on venetoclax in combination with carfilzomib and dexamethasone with improvement of κ free light chain levels. He underwent stem cell harvest with cyclophosphamide-based stem cell mobilization and subsequently received tandem ASCT with MEL200 conditioning for both transplants, achieving MRD-negative status.

Given his highly refractory disease at presentation, he was initially maintained on a venetoclax-based regimen, which he could not tolerate due to diarrhea. He was started on maintenance teclistamab and was continued on it on monthly basis for 2 years. Serial BM biopsies confirmed sustained

MRD-negative status throughout treatment. Teclistamab was discontinued due to recurrent infections. He remains in sustained MRD-negative remission at 12 months after the discontinuation of teclistamab.

This case illustrates key principles in managing primary refractory pPCL. Despite initial resistance to high-dose chemotherapy and quadruplet therapy, venetoclax-based treatment achieved cytoreduction, highlighting BCL-2 inhibition's potential in aggressive disease with t(11;14). The achievement of MRD-negative status with tandem ASCT demonstrates that even primary refractory pPCL can be controlled with intensive therapy following initial cytoreduction. Maintenance with teclistamab after tandem ASCT represents a novel strategy for sustaining MRD-negative remissions, with sustained negativity 12 months after maintenance suggesting the possibility of durable remissions in this challenging population using novel immunotherapies.

Given the aggressive biology and rapid proliferation characteristic of pPCL, we strongly recommend early use of the most potent available therapies rather than sequential conventional salvage approaches. Real-world studies suggest that although EMD and high-risk cytogenetics (frequently present in pPCL) may affect outcomes, BCMA-directed therapies including CAR T-cell therapy, remain highly active.^{73,78} Similarly, BCMA-targeted BsAbs such as teclistamab and G protein-coupled receptor class C group 5 member D (GPRC5D)-targeted BsAbs including talquetamab have shown high single-agent activity and should be considered early in the relapse setting.^{79,80} The advantage of BsAbs includes their off-the-shelf availability and potential for rapid initiation without the manufacturing delays associated with CAR T-cell therapy, which may be particularly important in rapidly progressive pPCL.

For patients with t(11;14), venetoclax-based therapy represents a rational treatment option at first relapse or suboptimal initial response.^{46,81-85} Multiple case reports and small series have demonstrated impressive responses to venetoclax, either as monotherapy or in combination with dexamethasone and daratumumab, in patients with relapsed pPCL harboring t(11;14). The presence of this translocation should be specifically evaluated at diagnosis to guide therapy selection at relapse. For patients without access to CAR T-cell therapy, BsAbs, or venetoclax (in t(11;14) cases), salvage chemotherapy combinations incorporating drugs not used in frontline therapy may be considered, although responses tend to be short-lived. Enrollment in clinical trials should be strongly encouraged given the critical unmet need for effective therapies in this population.

Management of sPCL

The management of sPCL requires highly individualized decision-making that carefully balances the goals of disease control against quality of life, patient performance status, and individual values. Unlike pPCL, in which aggressive up-front therapy can achieve prolonged remissions, patients with sPCL have already been exposed to multiple lines of therapy, often have accumulated treatment-related toxicities, experience significant comorbidities, and have profound treatment

resistance. The approach must therefore be personalized, considering the patient's treatment history, remaining therapeutic options, performance status, organ function, and most importantly, the patient's goals of care.⁸⁴⁻⁸⁹

For patients with preserved performance status and limited prior therapy exposure, cytoreductive approaches using available novel agents not previously used may provide symptom control and modest survival extension. T-cell redirecting therapies, including BCMA-directed CAR T cells or BsAbs, represent the most potent options for patients who have not been exposed to these modalities and maintain adequate organ function. However, the outcomes of those novel therapies in sPCL are suboptimal.⁸⁹⁻⁹² For patients with t(11;14), venetoclax-based combinations may offer meaningful responses. However, clinicians must recognize that response rates and durations in sPCL are significantly inferior to those observed in earlier-stage relapsed/refractory myeloma. For patients with poor performance status, multiple prior treatment exposures, significant organ dysfunction, or rapid disease progression despite recent therapy, early integration of palliative care, and discussions about comfort-focused approaches are essential. The key principle in sPCL management is that treatment decisions cannot follow a one-size-fits-all algorithm but must be tailored to each patient's unique clinical circumstances, remaining treatment options, and personal goals.

Future directions

The future for patients with PCL is promising, but realizing this potential requires a fundamental shift in approach including dedicated prospective trials, innovative strategies to prevent antigen escape through sequential targeting, and systematic inclusion of patients with PCL in clinical research.⁹³⁻⁹⁵ Patients with PCL are largely excluded from MM trials, leaving treatment decisions reliant on retrospective data and expert opinion. Future MM trials should mandate PCL inclusion with prespecified subgroup analyses, and dedicated prospective trials should be prioritized. Key strategies warranting prospective evaluation are as follows: (1) earlier immune-based therapies (BsAbs or CAR T cells) after ASCT to achieve durable MRD negativity; (2) MRD-guided treatment adaptation; and (3) novel combinations targeting BCL-2 dependence, immune evasion, and genomic instability. Sustained research commitment is essential to develop evidence-based frameworks for

this rare, high-attrition disease. The time for therapeutic nihilism in PCL has passed. With the remarkable array of effective agents now available and the existence of long-term survivors proving that durable disease control is achievable, the field has both the tools and responsibility to transform outcomes for patients with this devastating disease.

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Authorship

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Footnote

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REFERENCES

- van de Donk NW, Lokhorst HM, Anderson KC, Richardson PG. How I treat plasma cell leukemia. *Blood*. 2012;120(12):2376-2389.
- Gluzinski A, Reichenstein M. Myeloma und leukaemia lymphatica plasmocellularis [Myeloma and lymphatic plasmacytic leukemia]. *Wien Klin Wochenschr*. 1906;12:336-339.
- International Myeloma Working Group. Criteria for the classification of monoclonal gammopathies, multiple myeloma and related disorders: a report of the International Myeloma Working Group. *Br J Haematol*. 2003;121(5):749-757.
- Pagano L, Valentini CG, De Stefano V, et al. Primary plasma cell leukemia: a retrospective multicenter study of 73 patients. *Ann Oncol*. 2011;22(7):1628-1635.
- Tiedemann RE, Gonzalez-Paz N, Kyle RA, et al. Genetic aberrations and survival in plasma cell leukemia. *Leukemia*. 2008;22(5):1044-1052.
- Kyle RA, Maldonado JE, Bayrd ED. Plasma cell leukemia. Report on 17 cases. *Arch Intern Med*. 1974;133(5):813-818.
- Ravi P, Kumar SK, Roeker L, et al. Revised diagnostic criteria for plasma cell leukemia: results of a Mayo Clinic study with comparison of outcomes to multiple myeloma. *Blood Cancer J*. 2018;8(12):116.
- Fernández de Larrea C, Kyle R, Rosiñol L, et al. Primary plasma cell leukemia: consensus definition by the International Myeloma Working Group according to peripheral blood plasma cell percentage. *Blood Cancer J*. 2021;11(12):192.
- Jelinek T, Bezdekova R, Zihala D, et al. More than 2% of circulating tumor plasma cells defines plasma cell leukemia-like multiple myeloma. *J Clin Oncol*. 2023;41(7):1383-1392.
- Gonsalves WI, Rajkumar SV, Go RS, et al. Trends in survival of patients with primary plasma cell leukemia: a population-based analysis. *Blood*. 2014;124(6):907-912.
- Singh S, Peshin S, Wertheim BC, et al. Outcomes and treatment patterns in primary

- and secondary plasma cell leukemia: insights from a large US cohort study. *Haematologica*. 2025;110(9):2129-2138.
12. García-Sanz R, Orfão A, González M, et al. Primary plasma cell leukemia: clinical, immunophenotypic, DNA ploidy, and cytogenetic characteristics. *Blood*. 1999; 93(3):1032-1037.
 13. Ramsingh G, Mehan P, Luo J, Vij R, Morgensztern D. Primary plasma cell leukemia: a surveillance, epidemiology, and end results database analysis between 1973 and 2004. *Cancer*. 2009;115(24):5734-5739.
 14. Hofste Op Bruinink D, Kuiper R, van Duin M, et al. Identification of high-risk multiple myeloma with a plasma cell leukemia-like transcriptomic profile. *J Clin Oncol*. 2022; 40(27):3132-3150.
 15. Bladé J, Beksac M, Caers J, et al. Extramedullary disease in multiple myeloma: a systematic literature review. *Blood Cancer J*. 2022;12(3):45.
 16. Usmani SZ, Heuck C, Mitchell A, et al. Extramedullary disease portends poor prognosis in multiple myeloma and is over-represented in high-risk disease even in the era of novel agents. *Haematologica*. 2012; 97(11):1761-1767.
 17. Johnson MR, Del Carpio-Jayo D, Lin P, et al. Primary plasma cell leukemia: morphologic, immunophenotypic, and cytogenetic features of 4 cases treated with chemotherapy and stem cell transplantation. *Ann Diagn Pathol*. 2006;10(5):263-268.
 18. El Hussein S, Medeiros LJ, Hu S, Lin P, Wang W. The many faces of plasma cell neoplasms: morphological and immunophenotypical variants of the great imitator. *Pathology*. 2022;54(1):32-42.
 19. Katodritou E, Dalampira D, Delimpasi S, et al. Central nervous system multiple myeloma: a real-world multi-institutional study of the Greek Myeloma Study Group. *Am J Hematol*. 2024;99(10):1897-1905.
 20. Tuazon SA, Holmberg LA, Nadeem O, Richardson PG. A clinical perspective on plasma cell leukemia; current status and future directions. *Blood Cancer J*. 2021;11(2):23.
 21. Rojas EA, Corchete LA, Mateos MV, García-Sanz R, Misiewicz-Krzeminska I, Gutiérrez NC. Transcriptome analysis reveals significant differences between primary plasma cell leukemia and multiple myeloma even when sharing a similar genetic background. *Blood Cancer J*. 2019;9(12):90.
 22. Hellou T, Kumar SK, Alouch SS, et al. Impact of t(11;14) primary cytogenetic abnormality and the cumulative effect of multiple high-risk cytogenetics at diagnosis on the outcomes of patients with primary plasma cell leukemia. *Blood Cancer J*. 2025;15(1): 215.
 23. Guan J, Ma J, Chen B. Clinical and cytogenetic characteristics of primary and secondary plasma cell leukemia under the new IMWG definition criteria: a retrospective study. *Hematology*. 2023;28(1):2254556.
 24. Ramberger E, Sapozhnikova V, Ng YLD, et al. The proteogenomic landscape of multiple myeloma reveals insights into disease biology and therapeutic opportunities. *Nat Cancer*. 2024;5(8):1267-1284.
 25. Usmani SZ, Nair B, Qu P, et al. Primary plasma cell leukemia: clinical and laboratory presentation, gene-expression profiling and clinical outcome with Total Therapy protocols. *Leukemia*. 2012;26(11): 2398-2405.
 26. Tiedemann RE, Zhu YX, Schmidt J, et al. Identification of molecular vulnerabilities in human multiple myeloma cells by RNA interference lethality screening of the druggable genome. *Cancer Res*. 2012;72(3): 757-768.
 27. Lionetti M, Barbieri M, Todoerti K, et al. A compendium of DIS3 mutations and associated transcriptional signatures in plasma cell dyscrasias. *Oncotarget*. 2015; 6(28):26129-26141.
 28. Chapman MA, Lawrence MS, Keats JJ, et al. Initial genome sequencing and analysis of multiple myeloma. *Nature*. 2011;471(7339): 467-472.
 29. Bolli N, Avet-Loiseau H, Wedge DC, et al. Heterogeneity of genomic evolution and mutational profiles in multiple myeloma. *Nat Commun*. 2014;5:2997.
 30. Mithraprabhu S, Khong T, Ramachandran M, et al. Circulating tumour DNA analysis demonstrates spatial mutational heterogeneity that coincides with disease relapse in myeloma. *Leukemia*. 2017;31(8): 1695-1705.
 31. Lionetti M, Musto P, Di Martino MT, et al. Biological and clinical relevance of miRNA expression signatures in primary plasma cell leukemia. *Clin Cancer Res*. 2013;19(12): 3130-3142.
 32. Pichiorri F, Suh SS, Ladetto M, et al. MicroRNAs regulate critical genes associated with multiple myeloma pathogenesis. *Proc Natl Acad Sci U S A*. 2008;105(35): 12885-12890.
 33. Fernández de Larrea C, Kyle RA, Durie BG, et al. Plasma cell leukemia: consensus statement on diagnostic requirements, response criteria and treatment recommendations by the International Myeloma Working Group. *Leukemia*. 2013; 27(4):780-791.
 34. Van Riet I, De Waele M, Remels L, Lacom P, Schots R, Van Camp B. Expression of cytoadhesion molecules (CD56, CD54, CD18 and CD29) by myeloma plasma cells. *Br J Haematol*. 1991;79(3):421-427.
 35. Pellat-Deceunynck C, Bataille R, Robillard N, et al. Expression of CD28 and CD40 in human myeloma cells: a comparative study with normal plasma cells. *Blood*. 1994;84(8): 2597-2603.
 36. Cho H, Chung H, Kim JS, et al. Single-cell RNA sequencing identifies distinct immune microenvironment in patients with primary plasma cell leukemia [abstract]. *Blood*. 2023; 142(suppl 1):6583.
 37. Schinke C, Boyle EM, Ashby C, et al. Genomic analysis of primary plasma cell leukemia reveals complex structural alterations and high-risk mutational patterns. *Blood Cancer J*. 2020;10(6):70.
 38. Jurczyszyn A, Radocha J, Davila J, et al. Prognostic indicators in primary plasma cell leukaemia: a multicentre retrospective study of 117 patients. *Br J Haematol*. 2018;180(6): 831-839.
 39. Cazaubiel T, Leleu X, Perrot A, et al. Primary plasma cell leukemias displaying t(11;14) have specific genomic, transcriptional, and clinical features. *Blood*. 2022;139(17): 2666-2672.
 40. Palumbo A, Bringhen S, Mateos MV, et al. Geriatric assessment predicts survival and toxicities in elderly myeloma patients: an International Myeloma Working Group report. *Blood*. 2015;125(13):2068-2074.
 41. Al-Abbadi M, Al Hadidi S. Tumor lysis syndrome: pathophysiology, risk factors, management strategies, and prognostic indicators. *JAPA Acad J*. 2023;1(3):1-9.
 42. Thalambedu N, Al Hadidi S. Thromboprophylaxis in multiple myeloma. *Leuk Lymphoma*. 2022;63(12):2807-2815.
 43. Hussain M, Khan F, Al Hadidi S. The use of bone-modifying agents in multiple myeloma. *Blood Rev*. 2023;57:100999.
 44. Zweegman S, van de Donk NWCJ. Supportive care in myeloma-when treating the clone alone is not enough. *Hematol Am Soc Hematol Educ Program*. 2024;2024(1): 569-581.
 45. Barlogie B, Anaissie E, van Rhee F, et al. Incorporating bortezomib into upfront treatment for multiple myeloma: early results of total therapy 3. *Br J Haematol*. 2007; 138(2):176-185.
 46. Al Hadidi S. Primary plasma cell leukemia. *JAPA Acad J*. 2023;1(1):1-2.
 47. Kaiser MF, Hall A, Walker K, et al. Daratumumab, cyclophosphamide, bortezomib, lenalidomide, and dexamethasone as induction and extended consolidation improves outcome in ultra-high-risk multiple myeloma. *J Clin Oncol*. 2023;41(23):3945-3955.
 48. Musto P, Simeon V, Martorelli MC, et al. Lenalidomide and low-dose dexamethasone for newly diagnosed primary plasma cell leukemia. *Leukemia*. 2014;28(1):222-225.
 49. Musto P, Rossini F, Gay F, et al. Efficacy and safety of bortezomib in patients with plasma cell leukemia. *Cancer*. 2007;109(11): 2285-2290.
 50. Royer B, Minvielle S, Diouf M, et al. Bortezomib, doxorubicin, cyclophosphamide, dexamethasone induction followed by stem cell transplantation for primary plasma cell leukemia: a prospective phase II study of the

- Intergroupe Francophone du Myélome. *J Clin Oncol*. 2016;34(18):2125-2132.
51. van de Donk NWCJ, Minnema MC, van der Holt B, et al. Treatment of primary plasma cell leukaemia with carfilzomib and lenalidomide-based therapy (EMN12/HOVON-129): final analysis of a non-randomised, multicentre, phase 2 study. *Lancet Oncol*. 2023;24(10):1119-1133.
 52. Katodritou E, Kastritis E, Dalampira D, et al. Improved survival of patients with primary plasma cell leukemia with VRd or daratumumab-based quadruplets: a multicenter study by the Greek Myeloma Study Group. *Am J Hematol*. 2023;98(5):730-738.
 53. Parrondo RD, Moustafa MA, Reeder C, et al. Efficacy of daratumumab-based regimens for the treatment of plasma cell leukemia. *Clin Lymphoma Myeloma Leuk*. 2021;21(5):355-360.
 54. D'Arena G, Valentini CG, Pietrantonio G, et al. Frontline chemotherapy with bortezomib-containing combinations improves response rate and survival in primary plasma cell leukemia: a retrospective study from GIMEMA Multiple Myeloma Working Party. *Ann Oncol*. 2012;23(6):1499-1502.
 55. Mahindra A, Kalaycio ME, Vela-Ojeda J, et al. Hematopoietic cell transplantation for primary plasma cell leukemia: results from the Center for International Blood and Marrow Transplant Research. *Leukemia*. 2012;26(5):1091-1097.
 56. Dhakal B, Patel S, Girmius S, et al. Hematopoietic cell transplantation utilization and outcomes for primary plasma cell leukemia in the current era. *Leukemia*. 2020;34(12):3338-3347.
 57. Usmani SZ, Hoering A, Ailawadhi S, et al. Bortezomib, lenalidomide, and dexamethasone with or without elotuzumab in patients with untreated, high-risk multiple myeloma (SWOG-1211): primary analysis of a randomised, phase 2 trial. *Lancet Haematol*. 2021;8(1):e45-e54.
 58. Sonneveld P, Dimopoulos MA, Boccadoro M, et al. Daratumumab, bortezomib, lenalidomide, and dexamethasone for multiple myeloma. *N Engl J Med*. 2024;390(4):301-313.
 59. Mai EK, Bertsch U, Pozek E, et al. Isatuximab, lenalidomide, bortezomib, and dexamethasone induction therapy for transplant-eligible newly diagnosed multiple myeloma: final part 1 analysis of the GMMG-HD7 trial. *J Clin Oncol*. 2025;43(11):1279-1288.
 60. Leyboldt LB, Tichy D, Besemer B, et al. Isatuximab, carfilzomib, lenalidomide, and dexamethasone for the treatment of high-risk newly diagnosed multiple myeloma. *J Clin Oncol*. 2024;42(1):26-37.
 61. Perrot A, Lambert J, Hulin C, et al. Measurable residual disease-guided therapy in newly diagnosed myeloma. *N Engl J Med*. 2025;393(5):425-437.
 62. Touzeau C, Perrot A, Hulin C, et al. Daratumumab, carfilzomib, lenalidomide, and dexamethasone with tandem transplant for high-risk newly diagnosed myeloma. *Blood*. 2024;143(20):2029-2036.
 63. Callander NS, Silbermann R, Kaufman JL, et al. Daratumumab-based quadruplet therapy for transplant-eligible newly diagnosed multiple myeloma with high cytogenetic risk. *Blood Cancer J*. 2024;14(1):69.
 64. van Rhee F, Thanendrarajan S, Schinke CD, et al. Early results of TOTAL therapy 7 (TT7): high response rates of newly diagnosed high risk myeloma to daratumumab [abstract]. *Blood*. 2019;134(suppl 1):4569.
 65. Costa BA, Costa TA, Pak K, et al. Comparative efficacy of carfilzomib, lenalidomide, and dexamethasone (KRd) versus bortezomib, lenalidomide, and dexamethasone (VRd) in newly-diagnosed multiple myeloma: a systematic review and meta-analysis. *Am J Hematol*. 2024;99(7):1411-1414.
 66. Al Hadidi S, Ababneh OE, Schinke CD, et al. Long-term follow-up of patients with multiple myeloma treated on earlier total therapy protocols: a secondary analysis of 3 clinical trials. *JAMA Oncol*. 2025;11(8):910-915.
 67. Ali MO, Al Hadidi S. High dose (conditioning) regimens used prior to autologous stem cell transplantation in multiple myeloma. *Transpl Cell Ther*. 2022;28(9):572-580.
 68. Costa LJ, Bahlis NJ, Perrot A, et al. Teclistamab plus daratumumab in relapsed or refractory multiple myeloma. *N Engl J Med*. 2026;394(8):739-752.
 69. Munshi NC, Avet-Loiseau H, Anderson KC, et al. A large meta-analysis establishes the role of MRD negativity in long-term survival outcomes in patients with multiple myeloma. *Blood Adv*. 2020;4(23):5988-5999.
 70. Saccaro S, Fonseca R, Veillon DM, et al. Primary plasma cell leukemia: report of 17 new cases treated with autologous or allogeneic stem-cell transplantation and review of the literature. *Am J Hematol*. 2005;78(4):288-294.
 71. Morris C, Iacobelli S, Gahrton G, et al. Has allogeneic transplantation a role in the management of plasma cell leukaemia? A study on behalf of the Myeloma Subcommittee of the Chronic Leukaemia Working Party of the EBMT. *Blood*. 2011;118(21):2008.
 72. Schmidt WM, Perera ND, Buadi FK, et al. Long-term outcomes of allogeneic stem cell transplant in multiple myeloma. *Blood Cancer J*. 2023;13(1):126.
 73. Al Hadidi S, Heslop HE, Brenner MK, Suzuki M. Bispecific antibodies and autologous chimeric antigen receptor T cell therapies for treatment of hematological malignancies. *Mol Ther*. 2024;32(8):2444-2460.
 74. Facon T, Dimopoulos MA, Leleu XP, et al. Isatuximab, bortezomib, lenalidomide, and dexamethasone for multiple myeloma. *N Engl J Med*. 2024;391(17):1597-1609.
 75. Usmani SZ, Facon T, Hungria V, et al. Daratumumab plus bortezomib, lenalidomide and dexamethasone for transplant-ineligible or transplant-deferred newly diagnosed multiple myeloma: the randomized phase 3 CEPHEUS trial. *Nat Med*. 2025;31(4):1195-1202.
 76. Facon T, Moreau P, Weisel K, et al. Daratumumab/lenalidomide/dexamethasone in transplant-ineligible newly diagnosed myeloma: MAIA long-term outcomes. *Leukemia*. 2025;39(4):942-950.
 77. Grant SJ, Freeman CL, Rosko AE. Treatment of older adult or frail patients with multiple myeloma. *Hematol Am Soc Hematol Educ Program*. 2021;2021(1):46-54.
 78. Shrivastava T, Van Rhee F, Al Hadidi S. Targeting B cell maturation antigen in patients with multiple myeloma: current perspectives. *Onco Targets Ther*. 2023;16:441-464.
 79. Moreau P, Garfall AL, van de Donk NWCJ, et al. Teclistamab in relapsed or refractory multiple myeloma. *N Engl J Med*. 2022;387(6):495-505.
 80. Chari A, Minnema MC, Berdeja JG, et al. Talquetamab, a T-cell-redirecting GPRC5D bispecific antibody for multiple myeloma. *N Engl J Med*. 2022;387(24):2232-2244.
 81. Gonsalves WI, Buadi FK, Kumar SK. Combination therapy incorporating Bcl-2 inhibition with venetoclax for the treatment of refractory primary plasma cell leukemia with t(11;14). *Eur J Haematol*. 2018;100(2):215-217.
 82. Jelinek T, Mihalyova J, Kascak M, et al. Single-agent venetoclax induces MRD-negative response in relapsed primary plasma cell leukemia with t(11;14). *Am J Hematol*. 2019;94(1):E35-E37.
 83. Chakraborty R, Bhutani D, Lentzsch S. How do we manage t(11;14) plasma cell disorders with venetoclax? *Br J Haematol*. 2022;199(1):31-39.
 84. Ataergin S, Arpacı F, Kaya A, Cetin T, Gunhan O. VAD combination chemotherapy followed by bortezomib may be an effective treatment in secondary plasma cell leukemia. *Am J Hematol*. 2006;81(12):987-988.
 85. Ali R, Beksac M, Ozkalemkas F, et al. Efficacy of bortezomib in combination chemotherapy on secondary plasma cell leukemia. *Leuk Lymphoma*. 2007;48(7):1426-1428.
 86. Gozzetti A, Musto P, Defina M, et al. Efficacy of bortezomib, lenalidomide and dexamethasone (VRD) in secondary plasma cell leukaemia. *Br J Haematol*. 2012;157(4):497-498.
 87. Kupsh A, Arnall J, Voorhees P. A successful case of venetoclax-based therapy in relapsed/refractory secondary plasma cell leukemia. *J Oncol Pharm Pract*. 2020;26(5):1274-1278.

88. Shalaby K, Azad F, Parker S, et al. Clinical characteristics and survival outcomes of patients with primary and secondary plasma cell leukemia according to the 2021 definition: a single center retrospective study. *Clin Lymphoma Myeloma Leuk*. 2025;25(1):67-75.
89. Shrestha A, Thanendrarajan S, Bag A, et al. The efficacy of teclistamab in patients with multiple myeloma and secondary plasma cell leukemia. *Blood Adv*. 2025;9(19):4806-4809.
90. Galarza Fortuna GM, Peres L, Nazarenko E, et al. Safety and efficacy of BCMA-directed chimeric antigen receptor T-cell therapy for the treatment of plasma cell leukemia. *Blood Adv*. 2025;9(23):6009-6018.
91. Gaballa M, Julian K, Afrough A, et al. Real world outcomes of bispecific T-cell engagers in plasma cell leukemia [abstract]. *Blood*. 2025;146(suppl 1):823.
92. Packard D, Kumar S, Hellou T, et al. Patterns of progression and clinical outcomes after first relapse in patients with primary plasma cell leukemia [abstract]. *Blood*. 2025;146(suppl 1):1036.
93. Xu J, Bai Q, Liu Y, et al. CART-ASCT-CART2 sandwich regimen as frontline therapy: a phase II trial in patients with primary plasma cell leukemia [abstract]. *Blood*. 2025;146(suppl 1):4067.
94. Musto P, Engelhardt M, van de Donk NWCJ, et al. European Myeloma Network Group review and consensus statement on primary plasma cell leukemia. *Ann Oncol*. 2025;36(4):361-374.
95. Yan W, Du C, Zou H, et al. BCMA CAR T-cell therapy in newly diagnosed primary plasma cell leukemia ineligible for transplantation: an open label, single-arm, phase 2 study (CAREMM-002) [abstract]. *Blood*. 2025;146(suppl 1):257.

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